

Algorithm: Machine Learning

Speedup: Varies

Implementation: [Classiq \(QSVM\)](#), [Classiq \(Autoencoder\)](#), [PennyLane](#)

Description: Machine learning encompasses a wide variety of computational problems and can be attacked by a wide variety of algorithmic techniques. This entry summarizes quantum algorithmic techniques for improved machine learning. Many of the quantum algorithms here are cross-listed under other headings. In [\[214,250,251,309,338,339,359,403\]](#), quantum algorithms for solving linear systems [\[104\]](#) are applied to speed up cluster-finding, principal component analysis, binary classification, training of neural networks, and various forms of regression, provided the data satisfies certain conditions. (See also [\[433\]](#) for subsequent improvements to quantum principal component analysis.) However, a number of quantum machine learning algorithms based on linear systems have subsequently been "dequantized". Specifically, Tang showed in [\[400, 401\]](#) that the problems of recommendation systems and principal component analysis solved by the quantum algorithms of [\[251,309\]](#) can in fact also be solved in polynomial time randomized classical algorithms. The works [\[222,487,488,489,490\]](#) give quantum algorithms for topological data analysis by persistent homology. A cluster-finding method not based on the linear systems algorithm of [\[104\]](#) is given in [\[336\]](#). The papers [\[192,195,344,345,346,348\]](#) explore the use of adiabatic optimization techniques for the training of classifiers. In [\[456\]](#) quantum Ising machines are used for the training of classical neural networks. In [\[221\]](#), a method is proposed for training Boltzmann machines by manipulating coherent quantum states with amplitudes proportional to the Boltzmann weights. Polynomial speedups can be obtained by applying Grover search and related techniques such as amplitude amplification to amenable subroutines of state of the art classical machine learning algorithms. See, for example [\[358,340,341,342,343\]](#). Other quantum machine learning algorithms not falling into one of the above categories include [\[337,349\]](#). Some limitations of quantum machine learning algorithms are nicely summarized in [\[246\]](#). Many other quantum query algorithms that extract hidden structure of the black-box function could be cast as machine learning algorithms. See for example [\[146,23,11,31,212\]](#). Query algorithms for learning the majority and "battleship" functions are given in [\[224\]](#). Large quantum advantages for learning from noisy oracles are given in [\[236,237\]](#). In [\[428\]](#) quantum kernel estimation is used to implement a support-vector classifier solving a learning problem that is provably as hard as discrete logarithm. Several recent review articles [\[299,332,333\]](#) and a book [\[331\]](#) are available which summarize the state of the field. There is a related body of work, not strictly within the standard setting of quantum algorithms, regarding quantum learning in the case that the data itself is quantum coherent. See e.g. [\[350,334,335,351,352,353,354,355,356,357\]](#).